

ThreatWatch CyberLab

Software Design Document (SDD)



Gynia Prawl

Xavier University of Louisiana

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1. Problem Statement

Many cybersecurity students and entry-level analysts have a hard time gaining real-world experience since they don't have access to realistic security environments. Enterprise security systems like Security Information and Event Management (SIEM) platforms and real Security Operations Center (SOC) infrastructures are typically hard to use, cost a lot of money, and are not easy for beginners to get to. Because of this, students may know a lot about cybersecurity in theory but not have any real-world experience with things like assessing alerts, responding to situations, and figuring out how defensive decisions affect things.

ThreatWatch CyberLab fills this gap by offering a lightweight, simulation-based environment for intrusion detection that lets users practice finding and dealing with cyber threats in a safe setting. The tool lets students practice their investigative and defensive abilities in a safe way by mimicking real-life attack situations.

2. User Personas

ThreatWatch CyberLab is designed for users with varying levels of cybersecurity knowledge who want to develop practical experience identifying and responding to cyber threats. The following personas represent typical users who would benefit from the system.

2.1 Persona 1 - Sekani Gray (Cybersecurity Intern)

Sekani Gray is a 23-year-old cybersecurity intern who recently earned a bachelor's degree in Cybersecurity. He has a strong foundation in security concepts such as networking, intrusion detection, and incident response, but limited real-world experience working inside a Security Operations Center (SOC).

As an intern, Sekani wants to build confidence by applying what he learned in school to realistic scenarios. He is motivated to improve his technical skills, better understand security alerts, and prepare for a full-time cybersecurity role.

Demographics

- **Age:** 23
- **Education:** Bachelor's degree in Cybersecurity
- **Job Role:** Cybersecurity Intern
- **Technical Skill Level:** Entry-level to intermediate

Goals

- Practice working with intrusion detection systems
- Learn how to interpret and prioritize security alerts
- Gain hands-on experience responding to cyber threats
- Build confidence before transitioning into a full-time role

Frustrations / Pain Points

- Most training is theoretical and not hands-on
- Real SOC tools can be overwhelming for beginners
- Limited opportunities to safely practice making mistakes
- Difficulty connecting classroom knowledge to real incidents

How Sekani Uses ThreatWatch CyberLab

Sekani utilizes ThreatWatch CyberLab as a hands-on training environment, where he can observe simulated cyberattacks and observe how an intrusion detection system responds. He reviews alerts, decides how to respond, and learns from feedback on his actions. This allows Sekani to practice real-world cybersecurity workflows without the risk of impacting a live system.

Why This Persona Matters

This persona represents an early-career cybersecurity professional who needs practical experience to bridge the gap between education and real-world work. Designing ThreatWatch CyberLab with Sekani in mind ensures the platform focuses on usability, learning, and realistic security workflows rather than unnecessary complexity.

2.2 Persona 2 - Christiana Love (Undergrad Student)

Christiana Love is a 20-year-old undergraduate Computer Science student who is interested in cybersecurity but has limited exposure to security concepts and tools. She has completed introductory programming and networking coursework but has not yet worked with intrusion detection systems, security logs, or incident response processes.

Christiana is curious about cybersecurity as a potential career path, but feels overwhelmed by the complexity of professional security tools. She wants a supportive, low-risk environment where she can learn how cyberattacks are detected and handled without needing advanced prior knowledge.

Demographics

- **Age:** 20
- **Education:** Undergraduate Computer Science major
- **Job Role:** Student (pre-internship)
- **Technical Skill Level:** Beginner

Goals

- Understand how cybersecurity works in real-world settings
- Learn basic intrusion detection and response concepts
- Gain hands-on experience without fear of breaking systems
- Build confidence before taking advanced security courses

Frustrations / Pain Points

- Security terminology and tools feel intimidating
- Learning resources often assume prior security knowledge
- Difficulty visualizing how attacks and defenses interact
- Fear of making mistakes while learning

How Christiana Uses ThreatWatch CyberLab

Christiana uses ThreatWatch CyberLab as an introductory training platform. She begins with guided scenarios that explain alerts in simple language and offer clear response options. Through

repeated exposure and feedback, Christiana learns to recognize common attack patterns and understand why certain responses are effective, helping her gradually build foundational cybersecurity skills.

Why This Persona Matters

Christiana represents beginner users who are exploring cybersecurity for the first time. Designing ThreatWatch CyberLab with Christiana in mind ensures the platform remains accessible, educational, and supportive, making cybersecurity concepts easier to understand for users without a strong technical background.

2.3 Persona 3 - Dr. Angel Ryder (Cybersecurity Instructor)

Dr. Angel Ryder is a cybersecurity instructor and professor who teaches undergraduate and early graduate-level security courses. She has a strong background in cybersecurity concepts and professional experience, but is primarily focused on helping students translate theory into real-world practice.

Dr. Ryder wants a safe, structured platform that allows students to experience intrusion detection and response workflows without the cost, complexity, or risk of enterprise security systems.

Demographics

- **Age:** 47
- **Education:** Master's in Cybersecurity
- **Job Role:** Cybersecurity Instructor / Trainer
- **Technical Skill Level:** Advanced

Goals

- Provide students with hands-on cybersecurity experience
- Demonstrate how intrusion detection systems work in practice
- Teach decision-making and response workflows, not just theory
- Evaluate student understanding through realistic scenarios

Frustrations / Pain Points

- Limited access to realistic SOC-style environments for teaching
- Enterprise tools are too complex or expensive for classroom use
- Traditional labs focus on exploitation rather than defense
- Difficulty assessing how well students understand response processes

How Dr. Angel Ryder Uses ThreatWatch CyberLab

Dr. Ryder uses ThreatWatch CyberLab as an instructional and training tool. She assigns scenarios to students, allows them to interact with simulated intrusions, and reviews performance summaries to assess how students interpret alerts and respond to incidents. The platform supports guided learning while allowing Dr. Ryder to scale hands-on practice across multiple students.

Why This Persona Matters

Dr. Ryder represents instructors and trainers who shape how cybersecurity is taught. Designing ThreatWatch CyberLab with her needs in mind ensures the platform supports structured learning, guided progression, and assessment, making it suitable for classroom and training environments.

3. Functional Requirements

The following functional requirements outline the primary functionalities and behaviors of the ThreatWatch CyberLab system. These requirements explain how users will use the platform and how the system will operate in simulated cybersecurity situations.

FR-01: Intrusion Type Support

The system shall simulate and detect up to three categories of intrusion events: network-based intrusion, host-based intrusion, and password-based intrusion.

Rules / Details

- Each intrusion event shall be assigned one category.
- The category shall be visible to the user during investigation.

Acceptance Criteria

- Given a session is active, when an intrusion is generated, then it is classified as one of the supported types.
- Given an alert is opened, then the intrusion type is displayed.

FR-02: Attack Notification

When a simulated intrusion event is triggered, the system shall generate and display an alert notification to the user.

Rules / Details

- Each alert shall include:
 - intrusion type
 - severity level
 - timestamp

- short description
- Alerts shall appear in the Active Alerts panel.

Acceptance Criteria

- Given an intrusion occurs, when it is detected, then an alert appears in the Active Alerts panel.
- Given an alert appears, then it includes type, severity, and timestamp.

FR-03: Training Profile Selection

Before starting a session, the system shall require the user to select a training profile and configure the session accordingly.

Rules / Details

- Available profiles shall include:
 - Sekani Gray (intermediate)
 - Christiana Love (beginner)
 - Dr. Angel Ryder (instructor)
- The system shall adjust guidance level based on the selected profile.
- The session shall not begin until a profile is selected.

Acceptance Criteria

- Given no profile is selected, when the user attempts to start a session, then the system prevents session start.
- Given a profile is selected, when the session begins, then the system applies the corresponding guidance level.

FR-04: Randomized Attack Triggering

During an active session, the system shall trigger intrusion events at random intervals.

Rules / Details

- Events shall not require manual input to occur.
- Timing between events shall vary.

Acceptance Criteria

- Given a session is active, when time progresses, then intrusion events occur without user action.

FR-05: Investigation Capability

When an alert is displayed, the system shall allow the user to investigate the intrusion event and view its details.

Rules / Details

- Investigation view shall display:
 - intrusion type
 - severity level
 - indicators
 - current status

Acceptance Criteria

- Given an alert exists, when the user selects Investigate, then the system displays detailed intrusion information.

FR-06: Response Capability

After investigating an intrusion, the system shall allow the user to select a response to the event.

Rules / Details

- Only one response may be selected at a time.
- A response must be selected before resolution.

Acceptance Criteria

- Given an incident is active, when the user selects a response, then the system processes the selection.

FR-07: Alert Dismissal

The system shall allow the user to dismiss an alert without resolving the intrusion.

Rules / Details

- Dismissed alerts shall be removed from the Active Alerts panel.
- Dismissal may impact performance tracking.

Acceptance Criteria

- Given an alert exists, when the user selects Dismiss, then the alert is removed from the active list.

FR-08: Multiple Response Options

For each intrusion event, the system shall present multiple response options.

Rules / Details

- At least three response options shall be displayed.

- Options shall vary depending on intrusion type.

Acceptance Criteria

- Given an intrusion event, when the response screen is shown, then multiple response options are available.

FR-09: Attack Escalation

If the user selects an incorrect response or fails to respond within a given time, the system shall escalate the intrusion event.

Rules / Details

- Escalation shall:
 - increase severity level
 - update intrusion description
- Additional alerts may be generated.

Acceptance Criteria

- Given an incorrect response is selected, when evaluation completes, then severity increases.
- Given no response is selected, when time expires, then escalation occurs.

FR-10: Outcome Display

After a response is evaluated, the system shall display the outcome of the user's decision.

Rules / Details

- Outcome shall indicate:
 - contained

- partially contained
- escalated

Acceptance Criteria

- Given a response is selected, when evaluation completes, then the system displays the result.

FR-11: Explanation Feedback

After displaying the outcome, the system shall provide an explanation describing why the selected response succeeded or failed.

Rules / Details

- Explanation shall be simplified for beginner profiles.
- Explanation shall be concise for advanced users.

Acceptance Criteria

- Given an outcome is shown, when the explanation appears, then it describes the reasoning behind the result.

FR-12: Guided Learning Support

The system shall provide guided instructional support during training sessions based on the selected profile.

Rules / Details

- Beginner profile shall include:
 - hints
 - simplified explanations

- Advanced profile shall include minimal guidance.

Acceptance Criteria

- Given a beginner profile is selected, when interacting with alerts, then guidance is visible.
- Given an advanced profile is selected, then guidance is reduced.

FR-13: Points Reward System

When the user selects a correct response, the system shall award points.

Rules / Details

- Points shall be awarded once per resolved incident.
- Partially correct responses may award fewer points.

Acceptance Criteria

- Given a correct response is selected, when evaluation completes, then points are added to the user's total.

FR-14: Progress Tracking

During a session, the system shall track and update user performance metrics.

Rules / Details

- Metrics shall include:
 - total incidents
 - correct responses
 - incorrect responses
 - total points

Acceptance Criteria

- Given an incident is completed, when results are processed, then progress metrics update.

FR-15: Streak Tracking

The system shall track consecutive correct responses and update a streak count.

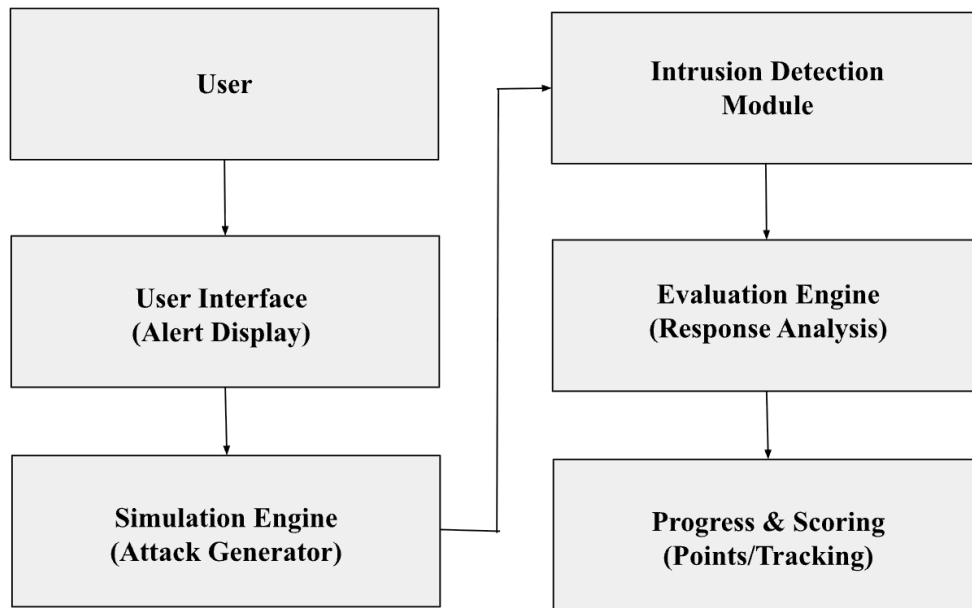
Rules / Details

- Streak increases with each correct response.
- Streak resets after an incorrect response.

Acceptance Criteria

- Given consecutive correct responses, when results are processed, then streak count increases.
- Given an incorrect response, when evaluated, then streak resets.

4. System Architecture



ThreatWatch CyberLab was constructed in a modular way, with many parts that work together. The user interface shows notifications and lets users look into and respond to fake attacks. This is how the user interacts with the system. The simulation engine makes up cyberattack scenarios that the intrusion detection module then works on. The system sends notifications to the user when it sees a threat. The user then has to look at the situation and decide how to respond. The evaluation engine looks at the user's answer and decides if the action worked. Finally, the progress and scoring module keeps track of how well users are doing, gives points for right answers, and keeps track of how far they've come in their training.

4.1 Intrusion Detection Module Design

The Intrusion Detection Module is responsible for analyzing simulated attack events generated by the Simulation Engine and determining whether the activity represents a cybersecurity threat. This module acts as the detection layer of the ThreatWatch CyberLab platform.

Input

The module receives simulated attack event data from the Simulation Engine. Each event contains structured information describing the simulated network activity. The input may include the following attributes:

- Attack type
- Source IP address
- Target system or port
- Number of connection attempts
- Timestamp of the activity
- Event identifier

Processing Steps

When an attack event is received, the Intrusion Detection Module performs the following steps:

1. The module receives the simulated attack event from the Simulation Engine.
2. The module parses the event data and extracts key attributes such as attack type, source IP address, and connection attempts.
3. The module compares the event characteristics against predefined detection rules.
4. If the event matches a known intrusion pattern, the module classifies the event as a detected intrusion.

5. The module assigns a severity level (Low, Medium, High) based on the characteristics of the attack.

Output

After detection and classification, the module generates an intrusion alert that is sent to the Evaluation Engine. The output includes:

- Intrusion type
- Severity level
- Source IP address
- Event identifier
- Detection timestamp

The Evaluation Engine then uses this information to present the alert to the user and evaluate the user's response to the simulated attack.

4.2 AIDE Integration (Detection Layer)

The ThreatWatch CyberLab system utilizes the open-source AIDE (Advanced Intrusion Detection Environment) configuration provided by Indiana University (2020) as the core detection layer. This component is responsible for identifying unauthorized changes to system files and generating data used to simulate intrusion events within the training environment.

AIDE Commands Used

The following commands are used to support intrusion detection:

- **Initialize database:** `aide --init`
- **Run integrity check:** `aide --check`
- **Update database after trusted changes:** `aide --update`

- **Filter detected changes:** `aide --check | grep "changed"`

Input

The Intrusion Detection Module receives:

- Simulated system file changes generated by the Simulation Engine
- File attributes including:
 - File name
 - File size
 - Permissions
 - Checksum values

Processing Steps

1. The system executes `aide --check` to compare current file states with the baseline database
2. AIDE analyzes file attributes and identifies discrepancies
3. Detected changes are parsed and categorized as potential intrusion events
4. The system maps detected changes to intrusion types:
 - File modification → Host-based intrusion
 - Unauthorized access attempts → Password-based intrusion
5. Relevant detection data is sent to the Evaluation Engine

Output

The Intrusion Detection Module produces:

- Alerts indicating detected anomalies
- File-level change reports (e.g., checksum mismatches)
- Structured event data used to:
 - Trigger user alerts
 - Populate investigation details
 - Evaluate user responses

Mapping to Functional Requirements

Functional Requirement	AIDE Command	Description
FR-01: Intrusion Type Support	aide --check	Detects file-based intrusions
FR-02: Attack Notification	aide --check	Generates data used for alerts
FR-05: Investigation Capability	aide --check	Provides file attribute details
FR-06: Response Capability	aide --check	Supplies data for decision-making
FR-09: Attack Escalation	aide --check	Identifies unresolved anomalies

5. Acceptance Criteria

Example 1

User story:

As a cybersecurity trainee, I want the system to evaluate the response I select for an intrusion alert so that I can learn whether my decision was correct.

Acceptance criteria:

- The system should evaluate the user's selected response after it is submitted.
- The system should determine whether the selected response is correct or incorrect based on the predefined scenario rules.
- The system should display the evaluation result to the user.
- The system should proceed to display the resulting state of the intrusion event after the response evaluation is completed.

Example 2

User story:

As a cybersecurity trainee, I want the system to escalate an intrusion event when I select an incorrect response so that I can understand the consequences of ineffective security decisions.

Acceptance criteria:

- If the user selects an incorrect response, the system should increase the severity level of the intrusion event.
- The system should simulate additional attack activity to represent escalation of the intrusion.

- The system should generate updated alerts reflecting the increased severity of the attack.
- The system should allow the user to continue responding to the escalated intrusion event.

References

Indiana-University. (2020, February 11). *Indiana-University/Puppet-aide: This puppet module manages the installation and configuration of aide (advance intrusion detection environment)*. GitHub. <https://github.com/indiana-university/puppet-aide>